



**ДЕПАРТАМЕНТ ОБРАЗОВАНИЯ И НАУКИ
ГОРОДА МОСКВЫ**
Государственное бюджетное профессиональное
образовательное учреждение города Москвы
«Колледж малого бизнеса № 4»
(ГБПОУ КМБ № 4)

Адаптивные технологии

Практическое занятие №7

Специальность: 09.02.07 Информационные системы и программирование

Форма обучения: очная

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Отчётная работа защищена с оценкой «___» _____

Москва, 2025 г.

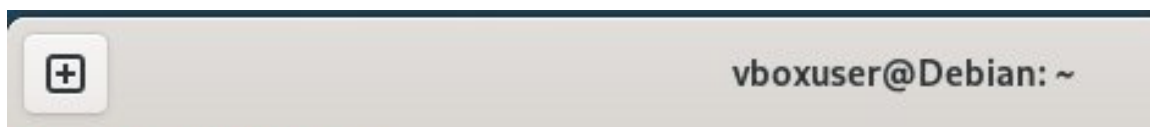
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Задание 1



Задание 2



```
vboxuser@Debian:~$ su
```

```
Password:
```

```
root@Debian:/home/vboxuser#
```

```
root@Debian:/home/vboxuser# lshw -C network
```

```
USB
```

```
SCSI
```

```
*-network
```

```
description: Ethernet interface
```

```
product: 82540EM Gigabit Ethernet Controller
```

```
vendor: Intel Corporation
```

```
physical id: 3
```

```
bus info: pci@0000:00:03.0
```

```
logical name: enp0s3
```

```
version: 02
```

```
serial: 08:00:27:01:42:3b
```

```
size: 1Gbit/s
```

```
capacity: 1Gbit/s
```

```
width: 32 bits
```

```
clock: 66MHz
```

```
capabilities: pm pcix bus_master cap_list ethernet physical tp 10bt 10bt-fd 100
```

```
fd 1000bt-fd autonegotiation
```

```
root@Debian:/home/vboxuser# lspci | grep -E "Ethernet|Network|Wireless"
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
root@Debian:/home/vboxuser#
```

```
user@DebianVM:~$ sudo ethtool enp0s3
Settings for enp0s3:
    Supported ports: [ TP ]
    Supported link modes:   10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full
    Supported pause frame use: No
    Supports auto-negotiation: Yes
    Supported FEC modes: Not reported
    Advertised link modes:  10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full
    Advertised pause frame use: No
    Advertised auto-negotiation: Yes
    Advertised FEC modes: Not reported
    Speed: 1000Mb/s
    Duplex: Full
    Auto-negotiation: on
    Port: Twisted Pair
    PHYAD: 0
    Transceiver: internal
    MDI-X: off (auto)
    Supports Wake-on: umbg
    Wake-on: d
    Current message level: 0x00000007 (7)
                           drv probe link

    Link detected: yes
```

```
user@DebianVM:~$
```

```
user@DebianVM:~$ sudo ethtool -i enp0s3
driver: e1000
version: 6.12.69+deb13-amd64
firmware-version:
expansion-rom-version:
bus-info: 0000:00:03.0
supports-statistics: yes
supports-test: yes
supports-eeprom-access: yes
supports-register-dump: yes
supports-priv-flags: no
```

```
user@DebianVM:~$
```

```
user@DebianVM:~$ sudo ethtool -P enp0s3
Permanent address: 08:00:27:05:c4:e2
user@DebianVM:~$
```

Задание 3

Cancel

Wired

Apply

DetailsIdentityIPv4IPv6Security

Link speed

1000 Mb/s

IPv4 Address

10.0.2.15

IPv6 Address

fd17:625c:f037:2:f8a7:dfb8:b5e3:748a
fd17:625c:f037:2:a00:27ff:fe01:423b
fe80::a00:27ff:fe01:423b

Hardware Address

08:00:27:01:42:3B

Default Route

10.0.2.2
fe80::2

DNS

10.166.186.13 10.89.87.63

☒ Connect automatically

☒ Make available to other users

☐ Metered connection: has data limits or can incur charges
Software updates and other large downloads will not be started automatically.

Remove Connection Profile...

Cancel

Wired

Apply

DetailsIdentityIPv4IPv6Security

IPv4 Method

☒ Automatic (DHCP)

☐ Link-Local Only

☐ Manual

☐ Disable

☐ Shared to other computers

DNS

Automatic

☒

Separate IP addresses with commas

Routes

Automatic

☒

Address	Netmask	Gateway	Metric

☐ Use this connection only for resources on its network

Cancel Wired Apply

Details Identity IPv4 **IPv6** Security

IPv6 Method

☒ Automatic ☐ Automatic, DHCP only

☐ Link-Local Only ☐ Manual

☐ Disable ☐ Shared to other devices

DNS Automatic ☒

Separate IP addresses with commas

Routes Automatic ☒

Address	Prefix	Gateway	Metric
			⊕

☐ Use this connection only for resources on its network

```
root@Debian:/home/vboxuser# ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
root@Debian:/home/vboxuser#
```

```
root@Debian:/home/vboxuser# ping fe80::a00:27ff:fe01:423b -c 5
ping: Warning: IPv6 link-local address on ICMP datagram socket may require ifname or scope-id =
> use: address%<ifname|scope-id>
PING fe80::a00:27ff:fe01:423b (fe80::a00:27ff:fe01:423b) 56 data bytes

--- fe80::a00:27ff:fe01:423b ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4091ms

root@Debian:/home/vboxuser#
```

```
user@DebianVM:~$ sudo route -n
Kernel IP routing table
Destination      Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0          10.0.2.2       0.0.0.0         UG    100    0      0 enp0s3
10.0.2.0         0.0.0.0       255.255.255.0   U     100    0      0 enp0s3
user@DebianVM:~$
```

```
user@DebianVM:~$ sudo netstat -rn
Kernel IP routing table
Destination      Gateway         Genmask         Flags  MSS  Window  irtt Iface
0.0.0.0          10.0.2.2       0.0.0.0         UG     0    0        0 enp0s3
10.0.2.0         0.0.0.0       255.255.255.0   U      0    0        0 enp0s3
user@DebianVM:~$
```

Задание 4

```
user@DebianVM:~$ ping 10.5.153.1 -c 5
PING 10.5.153.1 (10.5.153.1) 56(84) bytes of data.

--- 10.5.153.1 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4073ms
```

```
user@DebianVM:~$ ping ya.ru -c 5
PING ya.ru (213.180.193.56) 56(84) bytes of data.
64 bytes from familysearch.yandex.ru (213.180.193.56): icmp_seq=1 ttl=57 time=14.3 ms
64 bytes from familysearch.yandex.ru (213.180.193.56): icmp_seq=2 ttl=57 time=12.1 ms
64 bytes from familysearch.yandex.ru (213.180.193.56): icmp_seq=3 ttl=57 time=11.2 ms
64 bytes from familysearch.yandex.ru (213.180.193.56): icmp_seq=4 ttl=57 time=10.8 ms
64 bytes from familysearch.yandex.ru (213.180.193.56): icmp_seq=5 ttl=57 time=11.1 ms

--- ya.ru ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 10.760/11.887/14.306/1.295 ms
```

```
user@DebianVM:~$ traceroute -I ya.ru
traceroute to ya.ru (213.180.193.56), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.2) 0.609 ms 1.729 ms 1.741 ms
 2 gw-77.50.200-1.net.starlink.ru (77.50.200.1) 7.961 ms 7.955 ms *
 3 * * *
 4 100.127.255.18 (100.127.255.18) 7.627 ms * *
 5 * 100.127.255.13 (100.127.255.13) 10.753 ms *
 6 172.18.4.72 (172.18.4.72) 16.316 ms * *
 7 * * *
 8 * * 172.18.255.1 (172.18.255.1) 6.273 ms
 9 77.50.1.126 (77.50.1.126) 6.216 ms 6.188 ms 6.174 ms
10 * * *
11 172.18.52.113 (172.18.52.113) 8.336 ms * 7.744 ms
12 * * *
13 * * *
14 176.99.143.40.inetcom.ru (176.99.143.40) 5.748 ms * 7.987 ms
15 100.105.113.60 (100.105.113.60) 10.906 ms 14.085 ms *
16 m9-pl-ae21-915.yndx.net (93.158.160.240) 13.058 ms * *
17 * * *
18 10.4.1.1 (10.4.1.1) 13.776 ms * *
19 familysearch.yandex.ru (213.180.193.56) 12.880 ms * *
```

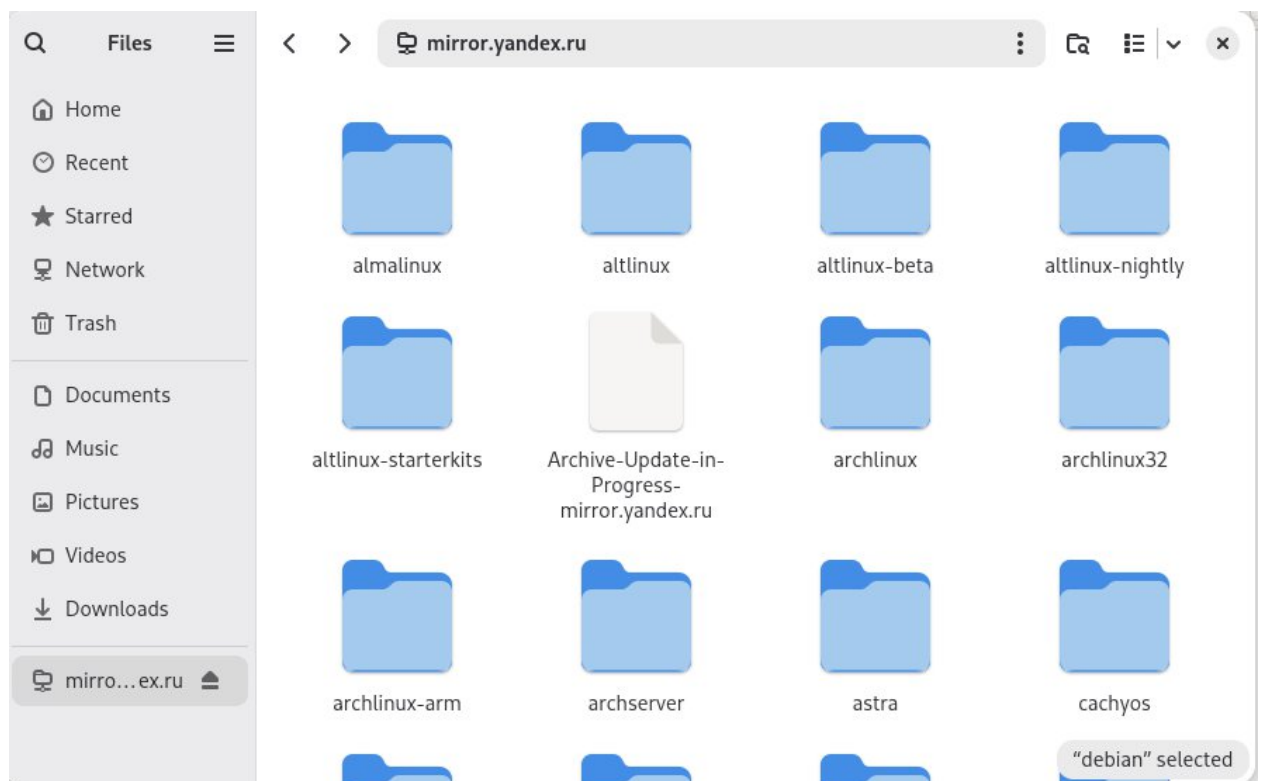
```
user@DebianVM:~$ tracepath ya.ru
1?: [LOCALHOST] pmtu 1500
1: _gateway 0.592ms
1: _gateway 0.702ms
2: no reply
3: no reply
4: no reply
5: no reply
6: no reply
```

Задание 5



Не получится на виртуальной машине

Задание 6



В файловом менеджере ввели адрес

Задание 7

```
user@DebianVM:~$ ssh root@brown.net.ru -p 65321
The authenticity of host '[brown.net.ru]:65321 ([82.146.54.86]:65321)' can't be established.
ED25519 key fingerprint is SHA256:GVogclgpZqWSUAQkP8uGM8lyRvp0te+QWII10Vw+eWA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[brown.net.ru]:65321' (ED25519) to the list of known hosts.
root@brown.net.ru's password:
Welcome to Ubuntu 16.04.7 LTS (GNU/Linux 4.4.0-210-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Feb 17 23:54:45 MSK 2026

System load:  0.0               Processes:            125
Usage of /:   72.8% of 39.24GB   Users logged in:     0
Memory usage: 55%              IP address for eth0: 82.146.54.86
Swap usage:   0%               IP address for ppp0: 176.16.10.1

Graph this data and manage this system at:
https://landscape.canonical.com/

23 packages can be updated.
6 updates are security updates.

New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Jan 12 17:50:40 2026 from 134.0.106.250
root@brown:~# █

Last login: Mon Jan 12 17:50:40 2026 from 134.0.106.250
root@brown:~# hostname
brown.net.ru
root@brown:~# █

root@brown:~# exit
logout
Connection to brown.net.ru closed.
user@DebianVM:~$ █
```

